Project Initialization and Planning Phase

Date	1July 2026
Team ID	xxxxxxx
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	The primary objective is to develop an intelligent credit card approval prediction system using machine learning techniques, enabling financial institutions to evaluate credit card applications quickly and accurately. The system aims to automate the approval process, reduce manual effort, minimize human errors, and assist decision-makers in identifying eligible and high-risk applicants.
Scope	The project focuses on automating the credit card approval process by analyzing applicants' financial and demographic information. It includes data collection, preprocessing, feature engineering, model training, evaluation of multiple machine learning algorithms, and integration of the best-performing model into a Flask web application for real-time prediction.
Problem Statement	
Description	Traditional credit card approval processes rely heavily on manual evaluation of applicant information, making the process time-consuming, inconsistent, and prone to human error. As the number of applications increases, financial institutions face challenges in making quick, accurate, and reliable approval decisions while effectively identifying high-risk applicants.
Impact	Automating the credit card approval process using machine learning improves operational efficiency, reduces processing time, enhances prediction accuracy, minimizes financial risks, and provides customers with faster eligibility assessments, ultimately improving customer satisfaction and organizational productivity.
Proposed Solution	

Approach	The proposed solution employs machine learning techniques to analyze applicant financial and demographic data and predict whether a credit card application is likely to be approved or rejected. Multiple classification algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, are trained and evaluated to identify the most accurate model. The selected model is integrated into a Flask-based web application, enabling users to receive real-time credit card approval predictions through an intuitive interface.
Key Features	• Implementation of multiple machine learning classification models for credit card approval prediction. • Automated data preprocessing and feature engineering for improved model performance. • Real-time prediction of credit card approval or rejection through a Flask web application. • High-risk applicant identification to support better credit risk assessment. • User-friendly interface for entering applicant information and viewing prediction results. • Accurate and consistent decision support for financial institutions.

Resource Requirements

Resource Type	**Description**	**Specification / Allocation Hardware**
Computing Resources	Processor	**Intel Core i3 or above**
Memory	RAM	**Minimum 4 GB (8 GB Recommended)**
Storage	Free Disk Space	**Minimum 2 GB**
Internet	Connectivity	**Required for dataset download, package installation, GitHub integration, and IBM Watson cloud deployment**
Display	Monitor	**Standard monitor with any modern web browser**
Software		
Framework	Backend Framework	**Flask**
Libraries	Python	**Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib,**

	Libraries	**Seaborn, Pickle**
Development Environment	IDE / Notebook	**Google Colab, Jupyter Notebook, Visual Studio Code**
Data	Source, Size, Format	**Kaggle Credit Card Approval Dataset (CSV format)**